

## Two open dMRI datasets (no credentials required)

Stanford HARDI — 10  $b = 0$  + 150 directions at  $b = 2000$  s/mm<sup>2</sup>  
Sherbrooke 3-shell — fitted subset: 1  $b = 0$  + 64 directions at  $b = 1000$  s/mm<sup>2</sup>

### held identical across every arm

one brain mask · one volume subset · no preprocessing  
*(MP-PCA denoising tested separately, applied once for all)*

#### FSL

dtifit  
--wls  
*measured*

#### MRtrix3

dwi2tensor  
-iter 0  
*measured*

#### MRtrix3

dwi2tensor  
(default)  
*predicted*

#### DIPY

WLS  
(default)  
*predicted*

#### FSL

dtifit  
(default)  
*unweighted*

#### DIPY

NLLS  
RESTORE  
*non-linear*

FA and MD maps, one set per arm

### physically admissible voxels only

FA in  $[0, 1]$      $0 < MD \leq 3.0 \times 10^{-3}$  mm<sup>2</sup>/s

Pearson  $r$  · mean absolute error  
Bland-Altman bias and 95% limits

## Tables 2-5, 8, 9 · Figure 2

### Phantoms with known eigenvalues

Rician noise at SNR 30, 20 and 10  
the five linear-fit arms

#### Table 6

### Simulations on the real gradient tables

idealised tensors, and each voxel's own fitted tensor  
Rician noise at the measured noise level

#### Table 7 · Tables S3-S5

All steps run inside one container: FSL 6.0.7 · MRtrix3 3.0.8 · DIPY 1.12.1